

Inferential modeling of food desert risk across U.S. census tracts

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NOTE: Using `readxl::read_xlsx()` with the original Excel file was not working for me, so I converted the sheet with the data to a CSV file and used `read.csv()`

```
# library(readxl)
library(relaimpo) # relative importance for the linear (forward stepwise) model
library(pls)      # for principal components regression (pcr)
library(corrplot)
library(caret)    # for train/test split and cross-validation utilities
library(glmnet)
library(tidyverse)
library(car)
library(vip)
```

Data Cleaning

```
#data <- read_xlsx("FoodAccessResearchAtlasData2019.xlsx", 3)
data <- read.csv("Food Access Research Atlas Data 2019.csv") # load data in
row.names(data) <- data$CensusTract # ID column
data <- select(data, -CensusTract) # remove duplicate
data <- replace(data, data == "NULL", NA) # change NULL cells to NA
# variables that need to be factors
flags <- c(
  "Urban",
  "GroupQuartersFlag",
  "LILATracts_1And10",
  "LILATracts_halfAnd10",
  "LILATracts_1And20",
  "LILATracts_Vehicle",
```

```

"HUNVFlag",
"LowIncomeTracts",
"LA1and10",
"LAhalfand10",
"LA1and20",
"LATracts_half",
"LATracts1",
"LATracts10",
"LATracts20",
"LATractsVehicle_20"
)

# variables that need to be numeric
numbers <- colnames(data)[c(4, 5, 7, 8, 15, 16, 25:length(data))]

data <- data |>
  mutate(
    across(all_of(flags), as.factor), # factorizing flags
    across(all_of(numbers), as.numeric), # numerizing numbers
    # low access share. if urban, >0.5 mi. if rural, >10 mi.
    lashare = ifelse(Urban == 1, lapophalf / Pop2010, lapop10 / Pop2010)
  )

# Only looking at low-income tracts for regression
data_li <- filter(data, LowIncomeTracts == 1)

# NOTE: If cells are NA, then the tract is not wide/long enough to be low-access
# at that threshold. Therefore, the tract technically has 0 low-access people
# at that threshold
data_regression <- replace(data_li, is.na(data_li), 0)

```

EDA

Summary statistics for the regression target and key predictors

```
summary(data_regression$lashare)
```

	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00000	0.02493	0.45722	0.45722	0.81964	1.00000	

```
summary(data_regression$MedianFamilyIncome)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0	38649	48203	48604	57321	250001

```
summary(data_regression$PovertyRate)
```

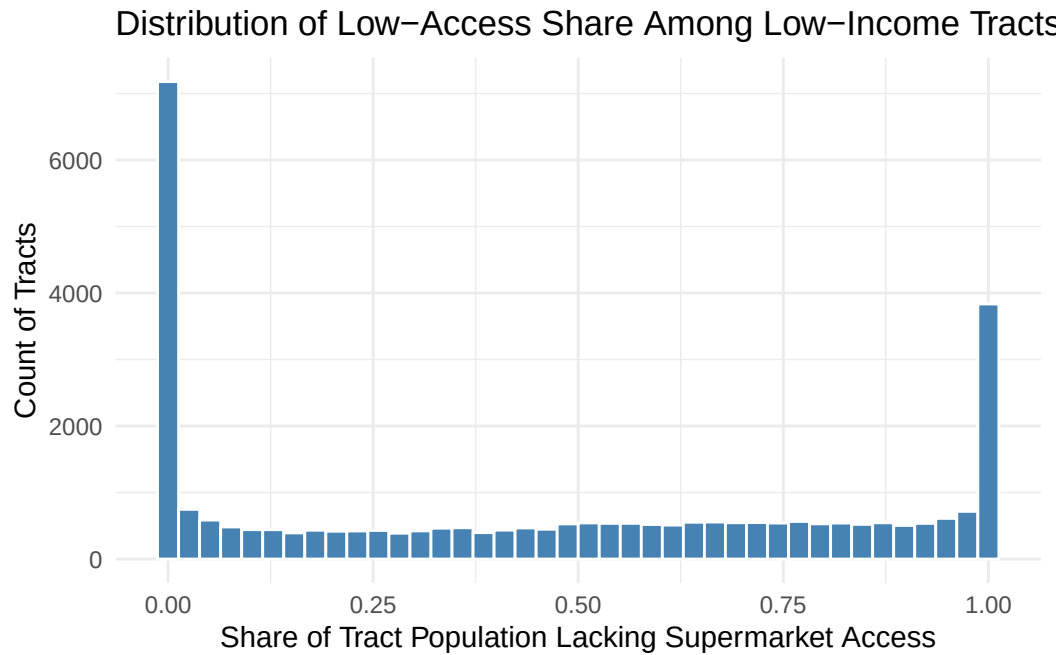
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.00	16.60	22.80	24.95	31.00	100.00

```
summary(data_regression$TractHUNV)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0	74.0	142.0	209.6	251.0	5449.0

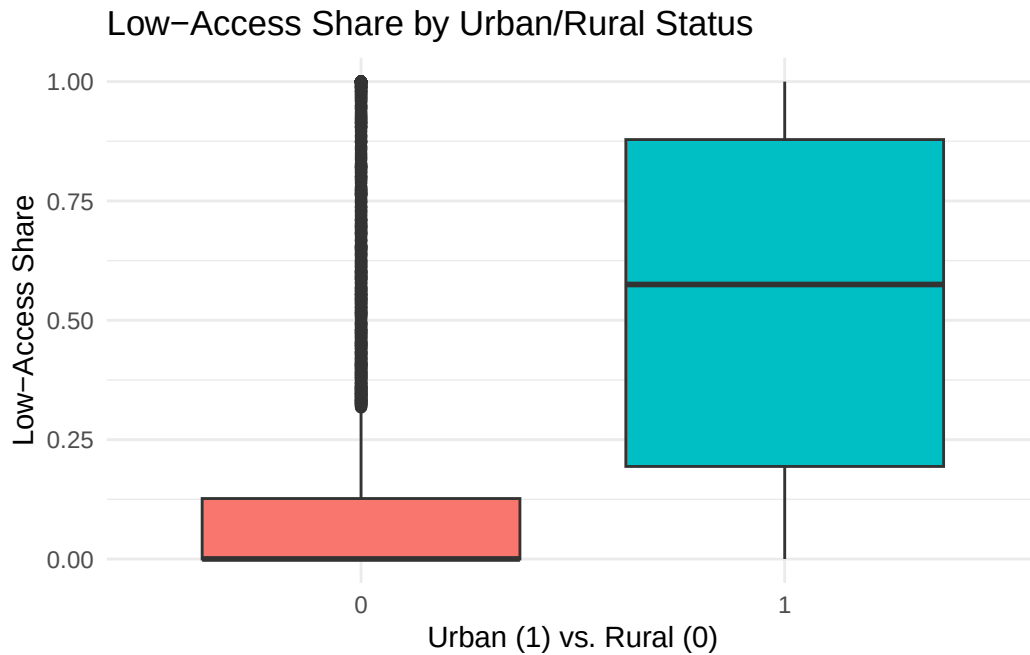
Distribution of the regression response

```
ggplot(data_regression, aes(x = lashare)) +  
  geom_histogram(bins = 40, fill = "steelblue", color = "white") +  
  labs(  
    title = "Distribution of Low-Access Share Among Low-Income Tracts",  
    x = "Share of Tract Population Lacking Supermarket Access",  
    y = "Count of Tracts"  
  ) +  
  theme_minimal()
```



Urban vs. rural comparison of low-access share

```
ggplot(data_regression, aes(x = Urban, y = lashare, fill = Urban)) +  
  geom_boxplot() +  
  labs(  
    title = "Low-Access Share by Urban/Rural Status",  
    x = "Urban (1) vs. Rural (0)",  
    y = "Low-Access Share"  
  ) +  
  theme_minimal() +  
  theme(legend.position = "none")
```

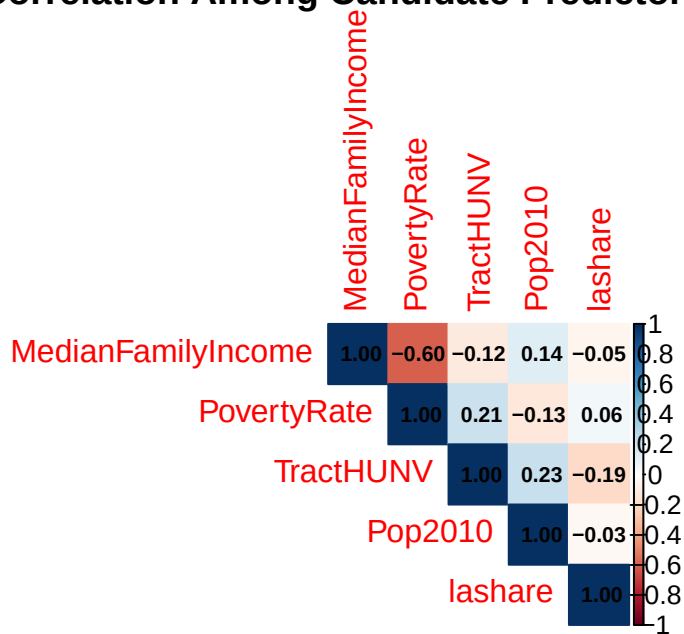


Correlation among candidate numeric predictors

```
# Selected to mirror the socioeconomic/geographic predictors named in the
# planned approach (income, poverty, vehicle access, population density)
eda_vars <- data_regression |>
  select(MedianFamilyIncome, PovertyRate, TractHUNV, Pop2010, lashare) |>
  mutate(across(everything(), as.numeric))

corr_matrix <- cor(eda_vars, use = "complete.obs")
corrplot(corr_matrix, method = "color", type = "upper",
  addCoef.col = "black", number.cex = 0.7,
  title = "Correlation Among Candidate Predictors",
  mar = c(0, 0, 1, 0))
```

Correlation Among Candidate Predictors



Class balance check for the classification target

```
table(data_regression$LILATracts_1And10)
```

```

      0      1
20994  9293

```

```
prop.table(table(data_regression$LILATracts_1And10))
```

```

      0      1
0.6931687 0.3068313

```

REGRESSION MODELING

Forward stepwise regression vs. principal components regression

Response: lashare (proportion of low-income tract lacking supermarket access)

Train/test split

```
train_idx <- createDataPartition(data_regression$lashare, p = 0.8, list = FALSE)
train_set <- data_regression[train_idx, ]
test_set <- data_regression[-train_idx, ]

# Candidate explanatory variables: socioeconomic and geographic features
# consistent with the planned approach in Section 5c
reg_predictors <- c(
  "MedianFamilyIncome", "PovertyRate", "TractHUNV", "Pop2010",
  "TractSNAP", "TractSeniors", "TractKids", "TractWhite", "TractBlack",
  "TractAsian", "TractHispanic", "Urban"
)

reg_formula <- as.formula(
  paste("lashare ~", paste(reg_predictors, collapse = " + "))
)
```

Full model (upper scope for stepwise search)

```
full_model <- lm(reg_formula, data = train_set)
```

Null model (lower scope for stepwise search)

```
null_model <- lm(lashare ~ 1, data = train_set)
```

Forward stepwise regression (AIC-based selection)

```
forward_model <- step(
  null_model,
  scope = list(lower = null_model, upper = full_model),
  direction = "forward",
  trace = 0
)
summary(forward_model)
```

Call:

```
lm(formula = lashare ~ Urban + TractHUNV + TractHispanic + TractSNAP +
```

```
TractAsian + Pop2010 + MedianFamilyIncome + TractSeniors +
TractKids + PovertyRate + TractBlack + TractWhite, data = train_set)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.80837	-0.21556	-0.03298	0.25237	1.83488

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.240e-02	1.356e-02	3.865	0.000112 ***
Urban1	5.347e-01	5.563e-03	96.113	< 2e-16 ***
TractHUNV	-4.618e-04	1.088e-05	-42.431	< 2e-16 ***
TractHispanic	-5.786e-05	2.602e-06	-22.236	< 2e-16 ***
TractSNAP	1.045e-05	1.639e-05	0.638	0.523706
TractAsian	-1.087e-04	8.001e-06	-13.584	< 2e-16 ***
Pop2010	-5.146e-06	5.566e-06	-0.925	0.355150
MedianFamilyIncome	-7.945e-07	1.654e-07	-4.803	1.57e-06 ***
TractSeniors	9.126e-05	9.298e-06	9.815	< 2e-16 ***
TractKids	6.427e-05	9.187e-06	6.996	2.70e-12 ***
PovertyRate	1.164e-03	2.291e-04	5.083	3.75e-07 ***
TractBlack	2.369e-05	5.049e-06	4.692	2.72e-06 ***
TractWhite	1.879e-05	4.881e-06	3.849	0.000119 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3084 on 24218 degrees of freedom

Multiple R-squared: 0.3318, Adjusted R-squared: 0.3315

F-statistic: 1002 on 12 and 24218 DF, p-value: < 2.2e-16

```
# Predicted values and R-squared on the held-out test set
forward_pred <- predict(forward_model, newdata = test_set)
forward_r2 <- cor(forward_pred, test_set$lashare)^2
forward_rmse <- sqrt(mean((forward_pred - test_set$lashare)^2))
```

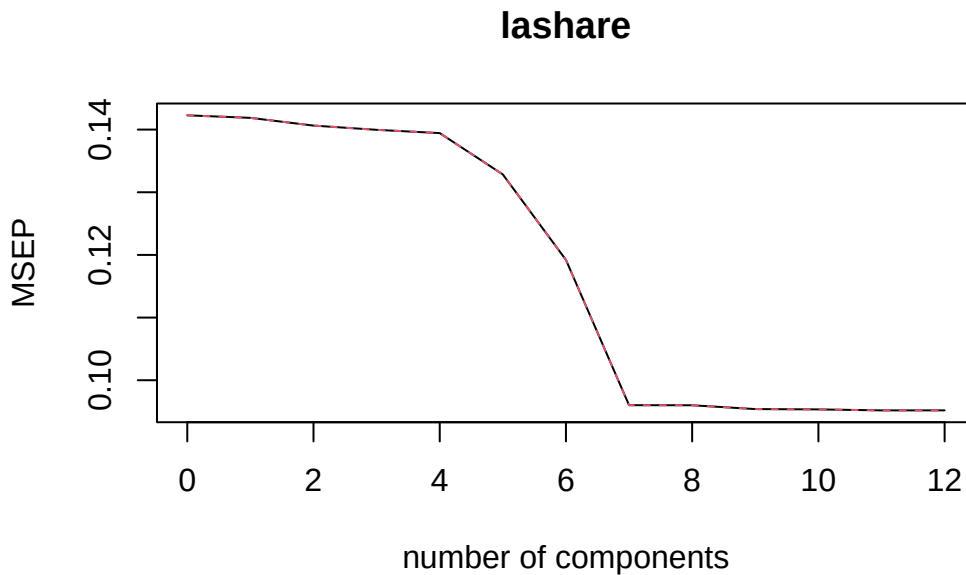
Principal Components Regression

```
pcr_model <- pcr(
  reg_formula,
  data = train_set,
  scale = TRUE,
  validation = "CV"
```



```
)

# Select number of components minimizing CV error
validationplot(pcr_model, val.type = "MSEP")
```



```
ncomp_best <- which.min(RMSEP(pcr_model)$val[1, 1, ]) - 1

pcr_pred <- predict(pcr_model, newdata = test_set, ncomp = ncomp_best)
pcr_r2 <- cor(as.vector(pcr_pred), test_set$lashare)^2
pcr_rmse <- sqrt(mean((as.vector(pcr_pred) - test_set$lashare)^2))
```

Model comparison: percentage of variance explained

```
model_comparison <- data.frame(
  Model = c("Forward Stepwise", "PCR"),
  R_squared = c(forward_r2, pcr_r2),
  RMSE = c(forward_rmse, pcr_rmse),
  N_Predictors = c(length(coef(forward_model)) - 1, ncomp_best[[1]])
)
print(model_comparison)
```

	Model	R_squared	RMSE	N_Predictors
1	Forward Stepwise	0.3315649	0.3080387	12
2	PCR	0.3313269	0.3080962	11

VARIABLE IMPORTANCE/INTERPRETABILITY ANALYSIS

Relative importance of predictors retained in the forward model

```
# lmg metric: decomposes R^2 by averaging over orderings (Lindeman, Merenda, Gold)
relimp <- calc.relimp(forward_model, type = "lmg", rela = TRUE)
print(relimp)
```

Response variable: lashare
 Total response variance: 0.1422704
 Analysis based on 24231 observations

12 Regressors:

Urban TractHUNV TractHispanic TractSNAP TractAsian Pop2010 MedianFamilyIncome TractSeniors T
 Proportion of variance explained by model: 33.18%
 Metrics are normalized to sum to 100% (rela=TRUE).

Relative importance metrics:

	lmg
Urban	0.671596587
TractHUNV	0.160475511
TractHispanic	0.045972738
TractSNAP	0.010159080
TractAsian	0.036131483
Pop2010	0.008685203
MedianFamilyIncome	0.004707524
TractSeniors	0.006418353
TractKids	0.008654614
PovertyRate	0.005926671
TractBlack	0.028658064
TractWhite	0.012614172

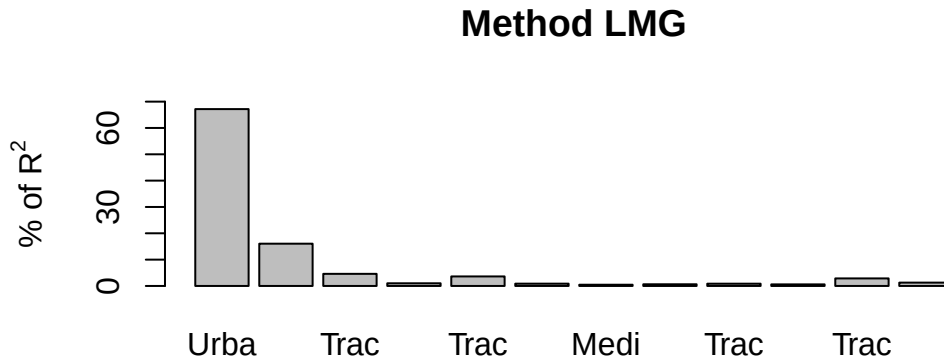
Average coefficients for different model sizes:

	1X	2Xs	3Xs	4Xs
Urban	4.046381e-01	4.191390e-01	4.344042e-01	4.500181e-01

TractHUNV	-2.852716e-04	-3.181463e-04	-3.500523e-04	-3.792067e-04
TractHispanic	-2.335174e-05	-2.621765e-05	-2.992241e-05	-3.398788e-05
TractSNAP	-2.776497e-05	-2.241057e-05	-2.002258e-05	-1.887167e-05
TractAsian	-1.152548e-04	-1.166054e-04	-1.180611e-04	-1.191070e-04
Pop2010	-7.139756e-06	-6.089942e-06	-5.303376e-06	-5.202473e-06
MedianFamilyIncome	-1.158933e-06	-1.033547e-06	-9.432575e-07	-8.883028e-07
TractSeniors	-4.735837e-05	-3.039083e-05	-1.397605e-05	9.077581e-07
TractKids	-3.263320e-06	8.913126e-06	1.862838e-05	2.607381e-05
PovertyRate	1.709454e-03	1.514070e-03	1.370279e-03	1.282330e-03
TractBlack	3.553233e-05	3.642548e-05	3.806133e-05	4.042054e-05
TractWhite	-7.667831e-06	-3.661191e-06	1.335017e-06	7.146387e-06
	5Xs	6Xs	7Xs	8Xs
Urban	4.654722e-01	4.801871e-01	4.936091e-01	5.053239e-01
TractHUNV	-4.046873e-04	-4.258630e-04	-4.421967e-04	-4.533317e-04
TractHispanic	-3.792547e-05	-4.135841e-05	-4.412790e-05	-4.633381e-05
TractSNAP	-1.781413e-05	-1.619302e-05	-1.379387e-05	-1.079308e-05
TractAsian	-1.190067e-04	-1.171874e-04	-1.135224e-04	-1.085244e-04
Pop2010	-6.082494e-06	-8.067241e-06	-1.102037e-05	-1.443947e-05
MedianFamilyIncome	-8.617878e-07	-8.541378e-07	-8.554463e-07	-8.569308e-07
TractSeniors	1.396356e-05	2.533065e-05	3.546969e-05	4.508751e-05
TractKids	3.172993e-05	3.637864e-05	4.096637e-05	4.631723e-05
PovertyRate	1.238682e-03	1.220811e-03	1.211388e-03	1.201139e-03
TractBlack	4.333098e-05	4.645628e-05	4.925091e-05	5.092884e-05
TractWhite	1.355317e-05	2.020879e-05	2.657574e-05	3.188686e-05
	9Xs	10Xs	11Xs	12Xs
Urban	5.151413e-01	5.231161e-01	5.294962e-01	5.346505e-01
TractHUNV	-4.593535e-04	-4.611199e-04	-4.606814e-04	-4.618007e-04
TractHispanic	-4.830437e-05	-5.052389e-05	-5.353689e-05	-5.786410e-05
TractSNAP	-7.588062e-06	-4.278231e-06	4.036381e-07	1.044885e-05
TractAsian	-1.034249e-04	-1.001249e-04	-1.010057e-04	-1.086863e-04
Pop2010	-1.736160e-05	-1.829113e-05	-1.514309e-05	-5.146365e-06
MedianFamilyIncome	-8.520972e-07	-8.375600e-07	-8.141513e-07	-7.944526e-07
TractSeniors	5.503473e-05	6.608930e-05	7.851055e-05	9.125722e-05
TractKids	5.275082e-05	5.963603e-05	6.486979e-05	6.427443e-05
PovertyRate	1.191090e-03	1.187551e-03	1.189118e-03	1.164366e-03
TractBlack	5.047530e-05	4.668946e-05	3.823852e-05	2.368956e-05
TractWhite	3.512983e-05	3.504945e-05	3.016467e-05	1.878913e-05

```
plot(relimp)
```

Relative importances for lashare



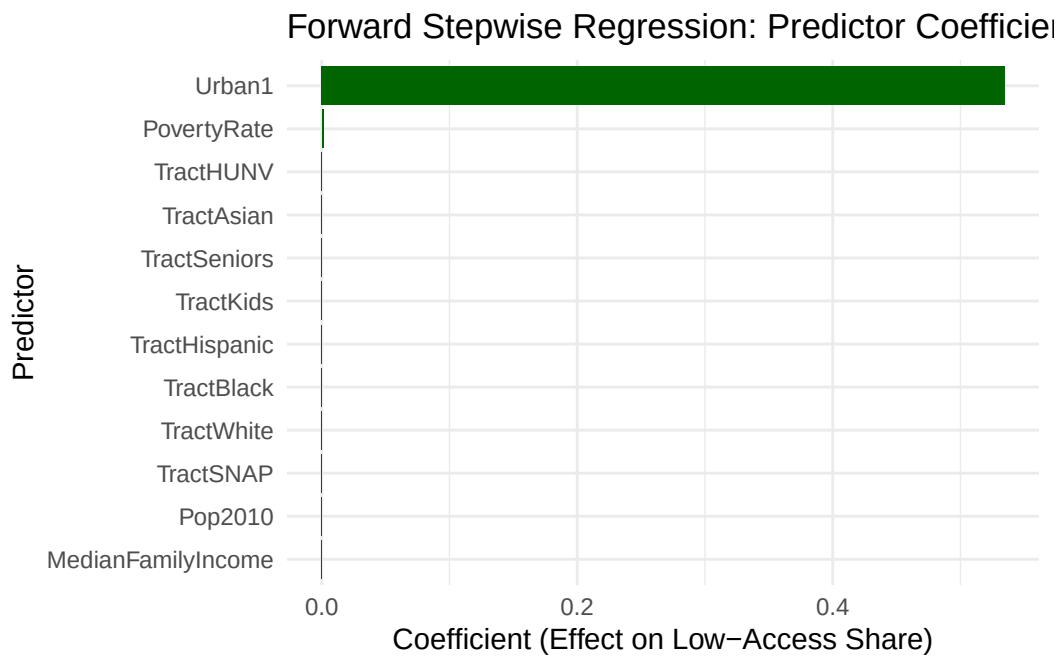
$R^2 = 33.18\%$, metrics are normalized to sum 100%.

Standardized coefficients for direct magnitude comparison

```
std_coefs <- data.frame(
  Predictor = names(coef(forward_model))[-1],
  Coefficient = coef(forward_model)[-1]
)
std_coefs <- std_coefs |> arrange(desc(abs(Coefficient)))
print(std_coefs)
```

	Predictor	Coefficient
Urban1	Urban1	5.346505e-01
PovertyRate	PovertyRate	1.164366e-03
TractHUNV	TractHUNV	-4.618007e-04
TractAsian	TractAsian	-1.086863e-04
TractSeniors	TractSeniors	9.125722e-05
TractKids	TractKids	6.427443e-05
TractHispanic	TractHispanic	-5.786410e-05
TractBlack	TractBlack	2.368956e-05
TractWhite	TractWhite	1.878913e-05
TractSNAP	TractSNAP	1.044885e-05
Pop2010	Pop2010	-5.146365e-06
MedianFamilyIncome	MedianFamilyIncome	-7.944526e-07

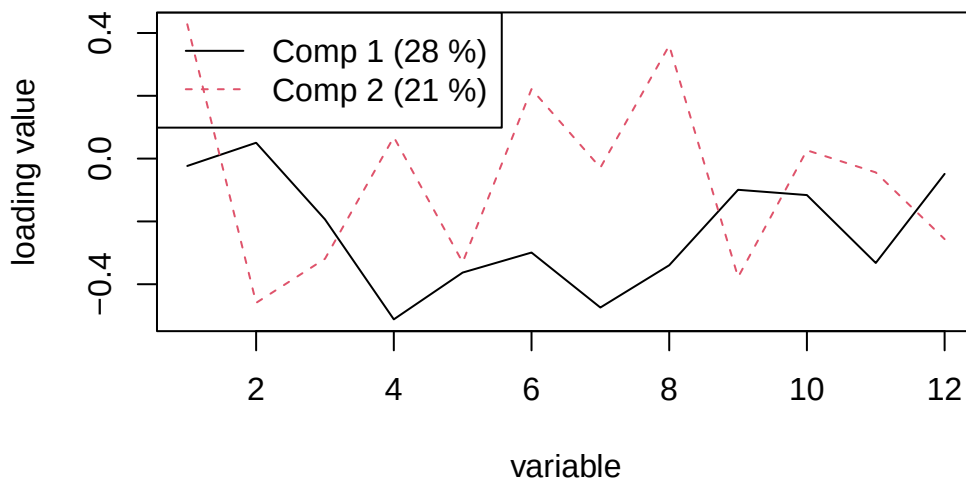
```
ggplot(
  std_coefs, aes(x = reorder(Predictor, abs(Coefficient)), y = Coefficient)
) +
  geom_col(fill = "darkgreen") +
  coord_flip() +
  labs(
    title = "Forward Stepwise Regression: Predictor Coefficients",
    x = "Predictor",
    y = "Coefficient (Effect on Low-Access Share)"
  ) +
  theme_minimal()
```



Loadings plot for PCR (which original variables drive top components)

```
loadingplot(pcr_model, comps = 1:2, legendpos = "topleft",
  main = "PCR Loadings: First Two Principal Components")
```

PCR Loadings: First Two Principal Components



Classification Modeling

LASSO

```
# share of occupied housing units with vehicles
data_classification <- data |>
  mutate(vehicle_share = 1 - (TractHUNV / OHU2010)) |>
  select(LILATracts_halfAnd10, PovertyRate, vehicle_share, Urban) |>
  drop_na()

set.seed(627)
train <- sample(nrow(data_classification), .8 * nrow(data_classification))
data_train <- data_classification[train, ]

# poverty rate -> PovertyRate
# vehicle availability -> vehicle_share
# population density -> Urban
# distance to nearest supermarket -> ???
train_mat <- model.matrix(glm(
  LILATracts_halfAnd10 ~ PovertyRate + vehicle_share + Urban,
  "binomial",
```

```
data_train
))
```

Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

```
data_test <- data_classification[-train, ]
test_mat <- model.matrix(glm(
  LILATracts_halfAnd10 ~ PovertyRate + vehicle_share + Urban,
  "binomial",
  data_test
))

set.seed(627)
data_lasso <- cv.glmnet(
  train_mat,
  data_train$LILATracts_halfAnd10,
  family = "binomial",
  type.measure = "auc"
)

best_t <- -1
best_conf_mat <- NULL
t_pos_rate <- c()
f_pos_rate <- c()
for (t in seq(0, 1, .01)) {
  conf_mat <- confusionMatrix(
    as.factor(
      ifelse(predict(data_lasso, newx = test_mat, type = "response") > t, 1, 0)
    ),
    data_test$LILATracts_halfAnd10,
    "1"
  )
  best_bal_acc <- ifelse(
    is.null(best_conf_mat),
    -1,
    best_conf_mat$byClass[["Balanced Accuracy"]]
  )
  t_pos_rate <- c(t_pos_rate, conf_mat$byClass[["Sensitivity"]])
  f_pos_rate <- c(f_pos_rate, 1 - conf_mat$byClass[["Specificity"]])
  # prioritize identifying LILA areas
  if (conf_mat$byClass[["Sensitivity"]] > .95)
    # do not want to overpredict LILA areas; main measure for comparison
```

```

    if (best_bal_acc < conf_mat$byClass[["Balanced Accuracy"]]) {
      best_conf_mat <- conf_mat
      best_t <- t
    }
  }
}

```

Warning in confusionMatrix.default(as.factor(ifelse(predict(data_lasso, :
Levels are not in the same order for reference and data. Refactoring data to
match.

Warning in confusionMatrix.default(as.factor(ifelse(predict(data_lasso, :
Levels are not in the same order for reference and data. Refactoring data to
match.

Warning in confusionMatrix.default(as.factor(ifelse(predict(data_lasso, :
Levels are not in the same order for reference and data. Refactoring data to
match.

Logit

```

data_logit <- glm(
  LILATracts_halfAnd10 ~ PovertyRate + vehicle_share + Urban,
  "binomial",
  data_train
)

```

Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

```
vif(data_logit) # multicollinearity check
```

PovertyRate	vehicle_share	Urban
1.671974	1.663960	1.088791

```

best_th <- -1
best_cm <- NULL
tpr <- c()
fpr <- c()
for (th in seq(0, 1, .01)) {
  cm <- confusionMatrix(
    as.factor(ifelse(predict(data_logit, data_test, "r") > th, 1, 0)),

```



```

    data_test$LILATracts_halfAnd10,
    "1"
  )
  best_ba <- ifelse(
    is.null(best_cm),
    -1,
    best_cm$byClass[["Balanced Accuracy"]]
  )
  tpr <- c(tpr, cm$byClass[["Sensitivity"]])
  fpr <- c(fpr, 1 - cm$byClass[["Specificity"]])
  if (cm$byClass[["Sensitivity"]] > .95)
    if (best_ba < cm$byClass[["Balanced Accuracy"]]) {
      best_cm <- cm
      best_th <- th
    }
}

```

Warning in confusionMatrix.default(as.factor(ifelse(predict(data_logit, :
Levels are not in the same order for reference and data. Refactoring data to
match.

Warning in confusionMatrix.default(as.factor(ifelse(predict(data_logit, :
Levels are not in the same order for reference and data. Refactoring data to
match.

Model Comparison

```

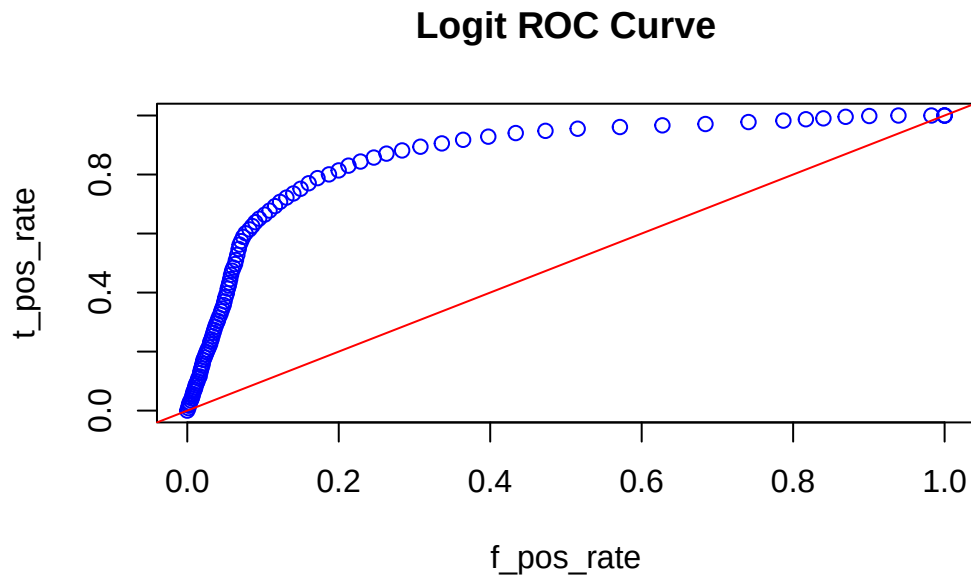
data.frame(
  "Model" = c("LASSO", "Logit"),
  "Threshold" = c(best_t, best_th),
  "Sensitivity" = c(
    best_conf_mat$byClass[["Sensitivity"]], best_cm$byClass[["Sensitivity"]]
  ),
  "Balanced Accuracy" = c(
    best_conf_mat$byClass[["Balanced Accuracy"]],
    best_cm$byClass[["Balanced Accuracy"]]
  )
)

```

	Model	Threshold	Sensitivity	Balanced.Accuracy
1	LASSO	0.14	0.9552127	0.7198670
2	Logit	0.10	0.9527246	0.6880477

ROC Curve

```
plot(f_pos_rate, t_pos_rate, main = "Logit ROC Curve", col = "blue")
abline(0, 1, col = "red")
```



Confusion Matrix

```
best_conf_mat
```

Confusion Matrix and Statistics

	Reference	
Prediction	0	1
0	5071	180
1	5395	3839

Accuracy : 0.6151
95% CI : (0.6071, 0.6231)
No Information Rate : 0.7225
P-Value [Acc > NIR] : 1

```
Kappa : 0.3142
McNemar's Test P-Value : <2e-16

Sensitivity : 0.9552
Specificity : 0.4845
Pos Pred Value : 0.4157
Neg Pred Value : 0.9657
Prevalence : 0.2775
Detection Rate : 0.2650
Detection Prevalence : 0.6375
Balanced Accuracy : 0.7199

'Positive' Class : 1
```

Variable Importance

```
coef(data_lasso)
```

```
5 x 1 sparse Matrix of class "dgCMatrix"
      lambda.1se
(Intercept) -5.1225917
(Intercept) .
PovertyRate  0.1083911
vehicle_share 1.3452255
Urban1      1.4236148
```

```
vip(data_lasso, mapping = aes())
```

